

Definition

Minimal Polynomial

Let T be a linear operator on a vector space. A polynomial $m(t)$ of least degree such that $m(T) = T_0$ is called the **minimal polynomial** if it is monic, meaning its coefficient of highest degree is 1.

Theorem

Uniqueness of Minimal Polynomials

Suppose T is a linear operator on a vector space with two minimal polynomials $m(t)$ and $m'(t)$ such that $m(T) = m'(T) = T_0$. Then $m(t) = m'(t)$.

Proof: Applying the division algorithm for polynomials, we get

$$m(t) = m'(t)q(t) + r(t).$$

Setting $t = T$ yields

$$m(T) = m'(T)q(T) + r(T).$$

Because $m(T)$ and $m'(T)$ are minimal polynomials, they send T to T_0 . We are then left with $r(T) = T_0$.

In general, $r(t) = 0$. Otherwise, it would be a polynomial, whose degree is less than the degree of $m'(t)$, that sends T to T_0 . That would be a contradiction. Hence, rewriting the equation for $m(t)$, we have

$$m(t) = m'(t)q(t).$$

Since $m(t)$ and $m'(t)$ are both characteristic polynomials, they have the same degree. Thus $\deg q(t) = 0$, so $q(t)$ is a constant function, meaning

$$m(t) = cm'(t).$$

Comparing coefficients of highest degree, we get $c = 1$. Therefore,

$$m(t) = m'(t). \quad \blacksquare$$

Proposition

$m(t)$ Divides $p(t)$

Let T be a linear operator on a complex finite-dimensional vector space. Then

$$m(t) \mid p(t).$$

Proof: Applying the division algorithm yields

$$p(t) = m(t)q(t) + r(t).$$

Substituting $t = T$, we get

$$p(T) = m(T)q(T) + r(T)$$

hence $r(T) = T_0$. Since $\deg r(t) < \deg m(t)$, it follows that $r(t) = 0$. Hence $m(t)$ divides $p(t)$. $\blacksquare$

Proposition

$m(t)$ and $p(t)$ Have the Same Roots

Suppose T is a linear operator on a complex finite-dimensional vector space V . Then $p(t)$ and $m(t)$ have exactly the same roots: the eigenvalues of T .

Proof: Let λ be an eigenvalue of T . Choose a corresponding nonzero eigenvector $v \in V$ such that $T(v) = \lambda v$.

The minimal polynomial $m(t)$ can be written as

$$m(t) = t^k + a_{k-1}t^{k-1} + \cdots + a_1t + a_0.$$

Setting $t = T$, we get

$$m(T) = T^k + a_{k-1}T^{k-1} + \cdots + a_1T + a_0I.$$

Hence $m(T)(v)$ expands to

$$\begin{aligned} m(T)(v) &= T^k(v) + a_{k-1}T^{k-1}(v) + \cdots + a_1T(v) + a_0v \\ &= \lambda^k v + a_{k-1}\lambda^{k-1}v + \cdots + a_1\lambda v + a_0v \\ &= (\lambda^k + a_{k-1}\lambda^{k-1} + \cdots + a_1\lambda + a_0)v \\ &= m(\lambda)v. \end{aligned}$$

However, since $m(T) = T_0$, we have $m(T)(v) = \mathbf{0}$. Hence $m(\lambda)v = \mathbf{0}$. Since v is nonzero, $m(\lambda) = 0$. Thus λ is a root of $m(t)$. Because λ was an arbitrary choice, it follows that the eigenvalues of T are the roots of $m(t)$, which we know is also true for $p(t)$. ■

Example

Let $T : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ be defined as

$$T(z_1, z_2, z_3) = (6z_1 + 3z_2 + 4z_3, 6z_2 + 2z_3, 7z_3).$$

Given the standard basis β for $\mathbb{C}$, the matrix representation for T under β is

$$[T]_{\beta} = \begin{pmatrix} 6 & 0 & 0 \\ 3 & 6 & 0 \\ 4 & 2 & 7 \end{pmatrix}.$$

We then find the characteristic polynomial:

$$p(t) = (t - 6)^2(t - 7)$$

Since $m(t)$ and $p(t)$ have the same roots, the only possibilities for $m(t)$ are

$$m(t) = p(t) \quad \text{or} \quad m(t) = (t - 6)(t - 7).$$

The second option has lower degree, so if it sends T to T_0 , it is the minimal polynomial. Testing this idea, we find

$$(T - 6I)(T - 7I) = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -3 & 0 & 0 \\ 6 & 0 & 0 \end{pmatrix} \neq 0_{3 \times 3}.$$

So the characteristic polynomial $p(t)$ is indeed the minimal polynomial $m(t)$.

Remark

Let T be a linear operator on an n -dimensional vector space V . Suppose T has n distinct eigenvalues. Then the characteristic polynomial $p(t)$ splits, meaning

$$p(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n).$$

Given that $m(t)$ and $p(t)$ have the same roots, the only possibility for $m(t)$ is

$$m(t) = p(t).$$

Definition

Jordan Form

A basis β for a vector space V is called a **Jordan basis** for a linear operator T on V if the matrix representation of T takes the form

$$[T]_{\beta} = \begin{pmatrix} A_1 & & \\ & A_2 & \\ & & \ddots \\ & & & A_m \end{pmatrix}$$

where each block $A_j = \lambda_j I$. We call each A_j a **Jordan block**.